

Self-Aware Networks

A Systems Whitepaper on Distributed Neural Rendering, Memory Scaling, and Embodied Action

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Abstract

The 2022 Self-Aware Networks (SAN) Whitepaper proposed Neural Array Projection Oscillation Tomography (NAPOT) as an organizing architecture for perception, memory, action, and artificial neurology. Its central intuition was that neural tissue does not deliver a complete sensory record to an inner viewer. Instead, successive neural populations receive partial signals, transform them according to learned cellular and circuit state, project changed signals into other populations, and participate collectively in the continuing construction of a body-and-world model. The same document proposed that structural memory changes alter what cells and circuits can recognize; that oscillatory relations help select, separate, and coordinate distributed activity; that recalled patterns can be reinstated at larger network scales; and that the distributed oscillating system performs the observer operation.

This whitepaper preserves those historical claims while restating them as an explicit, falsifiable systems model. We distinguish four levels of statement: established neuroscience, Micah Blumberg's historical interpretation, the current SAN hypothesis, and an engineering implication. NAPOT is formalized as a recurrent receive-transform-project architecture operating across cellular, local-circuit, and distributed-network scales. A literal tomography claim requires an identifiable hidden state, a specified forward projection, a reconstruction operator, held-out reconstruction advantage, and evidence that the reconstructed variable is used causally by the organism. Without those conditions, NAPOT remains a systems hypothesis rather than an established biological imaging process.

The current formulation replaces literal camera, microphone, screen, speaker, and projector language with measurable biological operations while preserving the original insight. Dendritic and somatic processes condition reception; membrane, synaptic, and circuit state transform incoming activity; axons and target-specific terminals project outputs; recurrent interactions alter distributed population trajectories; action changes the body and environment; and returned consequences update the next cycle. A bounded thalamocortical model specifies how repeated cortical-thalamic exchange could selectively sustain a task-relevant pattern, suppress competing noise, and return a concentrated signal without placing an observer in the thalamus. The observer is not a localized witness. The proposed observer operation is the active distributed system whose ongoing tonic state is changed by structured phasic differences and whose resulting state changes alter discrimination, report, memory, and action.

The paper compares SAN with dendritic integration, engram and attractor accounts, predictive processing, recurrent processing, global neuronal workspace models, reference-frame theories, communication-through-coherence, and dynamical-systems neuroscience. It proposes a nested experimental program using simultaneous cellular, mesoscopic, and systems measurements; perturbation and rescue; explicit comparator models; and separate endpoints for decoding, access, confidence, report, action, and phenomenology. Its contribution is to convert a public 2022 architecture into a bounded research program whose distinctive mechanisms can be measured, compared, supported, revised, or rejected.

Keywords: Self-Aware Networks; NAPOT; neural rendering; distributed observer; dendritic integration; oscillations; phase; thalamocortical recurrence; memory reinstatement; embodied action; consciousness; artificial neurology

1. Why a formal SAN whitepaper is needed

The original Whitepaper has an independent scholarly role. It is not merely an early version of one later SAN paper. It is the public systems map that placed several mechanisms into one sequence:

1. dendritic and cellular reception;
2. learned transformation by cellular and circuit state;
3. axonal projection into downstream receptive populations;
4. partial-cue completion and memory reinstatement;
5. distributed oscillatory coordination without an inner viewer;
6. embodied action and artificial implementation.

Later SAN papers can isolate NAPOT, Phase-Wave Differentials (PWD), the Field-Cell Self, choice, qualia, Artificial Neurology, or other mechanisms. SAN Book 2 can preserve the full history and broader argument. This whitepaper has a different job: it provides a concise map of how the components are meant to fit together and identifies the experiments needed to decide whether the resulting architecture adds explanatory value.

The modernization must not erase the historical record. It must also not repeat every 2022 sentence as established physiology. The paper therefore uses four labels:

- **Established:** a finding or constraint supported by cited neuroscience.
- **Historical SAN (2022):** a claim or analogy documented in the public 2022 Whitepaper.
- **Current SAN hypothesis:** the bounded mechanism proposed here.
- **Engineering implication:** a design possibility for artificial systems that does not establish biological truth or machine consciousness.

These labels are epistemic boundaries. They are not a ranking of importance. A SAN hypothesis can be central to the theory while remaining unconfirmed.

2. The architecture in ordinary language

2.1 The problem, proposed operation, and earned vocabulary

Turn your head while standing in a familiar room. The image at the eyes changes, yet the room normally remains stable. Reach for a light cup and the movement changes both the visual scene and the proprioceptive evidence returning from the arm. Recognize broccoli and the judgment joins shape, texture, color, prior experience, expected taste, bodily value, and an available action. These ordinary experiences expose the systems problem before any SAN term is introduced.

No single receptor, neuron, dendritic branch, cortical column, or sensory snapshot contains the complete body-and-world relation. The nervous system must combine partial signals that arrive at different places and times, transform them according to learned structure and current state, maintain useful relations while the organism moves, select an action, and use the returned consequence to correct the next estimate.

SAN proposes a repeated operation: partial evidence changes a receiver; that receiver transforms the event through its present and learned state; changed output reaches selected targets; recurrent exchange recruits a larger relational pattern; the pattern changes prediction or action; and the consequences return as new evidence. The reader can understand and test this operation without any coined terminology.

The nervous system receives incomplete, noisy, and body-dependent evidence. A receptor, neuron, or population does not possess the whole scene. Each receives only the signals that reach it through its current anatomical and functional connections.

What happens next depends on prior structure and current state. Dendritic branches, receptors, membrane conductances, inhibition, neuromodulation, synaptic history, and ongoing population activity alter how the same incoming event affects the receiver. The receiver transforms input rather than copying it.

Changed activity is then projected to selected downstream targets. Different axon branches and synapses can have different effects because the targets have different receptors, membrane states, circuit contexts, and histories. A neural output is therefore receiver-relative: the same presynaptic event need not have the same consequence everywhere.

Many local transformations occur concurrently. Recurrent and feedback pathways allow later activity to influence earlier or parallel processing. Learned assemblies can be recruited by partial cues. Distributed populations can settle into a useful state without reproducing every neuron or spike from a prior episode.

The resulting state matters only if it changes what the organism can do. It can alter recognition, expectation, valuation, confidence, action selection, movement, or report. Action changes the sensory evidence that returns to the nervous system, continuing the cycle.

In compressed form, the architecture is:

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receive partial evidence
-> transform it through learned and current state
-> project changed signals to selected targets
-> coordinate and complete distributed relations
-> alter prediction, discrimination, and action
-> receive the consequences of action
-> update the next cycle
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The continuing construction of an action-usable internal model is called **neural rendering**. When the receive-transform-project operation is evaluated across nested neural arrays, oscillatory relations, recurrent reconstruction, and returned action consequences, SAN calls the proposed architecture **Neural Array Projection Oscillation Tomography (NAPOT)**. These names summarize the operation already described; they do not substitute for it. A literal tomography claim has additional mathematical and causal requirements defined in Section 5.4.

3. Historical formulation and current formulation

The table below preserves the major 2022 claims without treating their original language as settled evidence.

2022 formulation	Current bounded formulation	What would support it	What would weaken it
A dendrite is like an eye or microphone.	Dendritic and somatic compartments receive and nonlinearly integrate spatially and temporally structured synaptic input.	Compartment-resolved recordings show that specific input relations add predictive or causal value beyond somatic rate alone.	The proposed dendritic variables add no held-out or intervention value.
An exit terminal is like a pixel or speaker.	Axons and target-specific terminals transmit activity whose consequences depend on release, receptor, dendritic location, and receiver state.	Target-specific output variables predict and causally alter downstream state.	A simpler rate-only model explains the same results across targets.

2022 formulation	Current bounded formulation	What would support it	What would weaken it
A synaptic memory scales to the whole brain.	Learned structural and physiological readiness can help a partial cue recruit a distributed state through recurrent pathways.	Cue-specific perturbation and rescue reveal a directed recruitment path and action-usable population reconstruction.	Recall requires no proposed route, or stimulation produces only nonspecific activation.
Oscillations bind the rendering.	Relative phase, coherence, structured dispersion, nested rhythms, and state-dependent coupling may regulate which distributed relations become effective.	Phase-aware models outperform power, rate, anatomy, and task covariates and survive selective perturbation-rescue.	Phase variables add no generalizable information or causal control.
The oscillator is the observer.	The observer operation is the distributed body-brain system whose ongoing state is changed by evidence and whose changed state alters later discrimination and action.	Multiscale state variables jointly explain content-specific behavior beyond any single local readout.	A local mechanism or non-oscillatory comparator explains the complete causal chain more parsimoniously.
The brain performs tomography.	Literal NAPOT requires a specified hidden state, forward projection, identifiable inverse, uncertainty estimate, and endogenous use of the reconstruction.	Held-out reconstruction and causal use exceed Bayesian, recurrent, attractor, and latent-state comparators.	The inverse is non-identifiable, decoder-only, or unused by the organism.
Artificial neural networks can become self-aware through NAPOT.	NAPOT motivates architectures with persistent state, receiver-relative transformation, recurrent reconstruction, embodiment, and consequence-driven updating.	Implementations outperform controls on explicit capabilities and pass separately defined consciousness-relevant tests.	Architectural similarity fails to produce the predicted capabilities; no consciousness inference follows from implementation alone.

4. The six-stage SAN architecture

4.1 Stage 1: reception

Established. Neurons receive thousands of synaptic inputs distributed across dendritic and somatic compartments. Dendritic branches can perform nonlinear operations, including local spikes and location-dependent integration. In rat layer 2/3 pyramidal neurons, apical dendritic spikes can influence somatic output [12]. In mouse primary-visual-cortex layer 5 pyramidal neurons, apical-basal coincidence can regulate bursting [18]. A separate CA1 anatomy-and-model study supports branchwise two-stage integration [11]. These findings do not imply that every neuron performs the same computation or that a dendrite stores a complete image.

Historical SAN (2022). The Whitepaper described the dendrite as a pattern-sensitive receiver, using the eye and microphone as explanatory analogies. It called the incoming dendritic domain a receptive field and emphasized that a receiving array samples outputs from many preceding neurons.

Current SAN hypothesis. The relevant unit is a receiver-conditioned event. Let neuron or population i receive input vector $x_i(t)$ under internal state $s_i(t)$, structural parameters m_i , and contextual modulation $c_i(t)$. Its reception is

$$r_i(t) = R_i(x_i(t), s_i(t), m_i, c_i(t)). \quad (1)$$

The function R_i is not assumed to be universal. It may include dendritic subunits, inhibitory gating, neuromodulation, membrane conductances, and recent history. The empirical question is which measured variables are necessary to predict or control the receiver’s next state.

Engineering implication. Artificial units should not be limited to a scalar weighted sum when the task requires context-dependent reception. A receiver can expose typed input channels, branch-like subunits, state-dependent thresholds, and explicit uncertainty.

4.2 Stage 2: transformation

Established. Synaptic efficacy, dendritic morphology, receptor composition, intrinsic excitability, inhibition, and neuromodulation can change how subsequent inputs affect a neuron or circuit. In a mouse contextual-fear amnesia model, specific engram-cell connectivity persisted and optogenetic ensemble reactivation could restore recall [17]. This demonstrates a causal connectivity-and-reactivation contribution in that preparation; it does not reduce long-term memory generally to one cell or structure.

Historical SAN (2022). The Whitepaper described changed synapses and dendritic morphology as a physical memory because they alter what patterns a cell can recognize. It called this learned pattern-sensitive structure an Expert Data Structure.

Current SAN hypothesis. Memory-bearing structure is *changed readiness*. It constrains future state transitions but does not contain a complete episode by itself. Transformation is written as

$$z_i(t) = T_i(r_i(t), s_i(t), m_i, c_i(t)), \quad (2)$$

and plastic updating as

$$m_i(t+1) = m_i(t) + U_i(r_i(t), z_i(t), \mathbf{y}_i^{\text{out}}(t), e_i(t)), \quad (3)$$

where the bold output term denotes the declared collection of target-specific outputs $y_{i \rightarrow j}(t)$ for the targets j of receiver i , and $e_i(t)$ denotes returned consequence or error-relevant evidence. The update U_i may include synaptic, intrinsic, structural, and circuit-level changes. SAN does not require all such changes to occur on the same timescale.

Engineering implication. A useful artificial analogue separates persistent learned structure from active state and records how an observation changed future routing or response readiness.

4.3 Stage 3: projection

Established. Neuronal outputs travel through axons to anatomically constrained targets. Macaque interareal projections are weighted and directed [13]. In two macaques performing an attention task, coherence, power-correlation, and Granger-causal interaction networks differed across rhythms and related to anatomy in frequency-dependent ways [19].

Historical SAN (2022). The Whitepaper used *inceptive field* for the branching target field created by a neuron’s axon and terminals. It compared one neuron’s output with one pixel or speaker contribution to a larger downstream pattern.

Current SAN hypothesis. The biologically accurate projection variable is axonal and synaptic output, not apical-dendritic broadcast. For source i and target j , define

$$y_{i \rightarrow j}(t) = P_{i \rightarrow j}(z_i(t), a_{ij}, q_{ij}(t), s_j(t)), \quad (4)$$

where a_{ij} is anatomical connectivity, q_{ij} represents terminal and release state, and s_j is the receiver state. The output event may include spike count, timing, phase relative to a local reference, burst structure, waveform features, and transmitter release. Which components matter is an experimental question.

Engineering implication. A message should be typed by destination and consequence rather than treated as a context-free token. The same source event can be transformed differently by different receivers.

4.4 Stage 4: completion and memory scaling

Established. In a mouse contextual-fear model of protein-synthesis-inhibition-induced amnesia, strong optogenetic activation of dentate-gyrus engram cells reinstated recall, and downstream engram connectivity was necessary [17]. This supports distributed connectivity and reactivation in that preparation; it does not alone establish general attractor dynamics, partial-cue completion, cortical reinstatement, or cellular-participation drift.

Historical SAN (2022). The Whitepaper proposed that a small memory-bearing structure can be magnified or scaled into a larger active pattern. It described oscillating groups, inhibition, sharp-wave ripples, and recurrent routes as candidate mechanisms.

Current SAN hypothesis. A local structure does not contain and replay a lossless whole-brain image. A cue recruits a distributed reconstruction whose content depends jointly on retained structure and present network state:

$$\hat{h}_t = G(c_t, M, s_t, A), \quad (5)$$

where c_t is the cue, M is distributed learned structure, s_t is current state, and A is the constrained anatomical and effective-connectivity graph. The reconstruction $\hat{h}_t$ may be useful without reproducing the original microstate.

A literal memory-scaling claim requires more than increased activity. It requires evidence that a bounded cue recruits a content-specific distributed state, that the state carries relations absent from the cue, and that perturbing the proposed route changes behavior or report in a predicted way.

Engineering implication. Memory systems should support partial-cue reconstruction through shared latent and relational structure rather than retrieve only a fixed record.

4.5 Stage 5: distributed rendering and observation

Established. Early feedforward visual activity can persist under masking even when later reentrant signatures and objective detection are lost [7]. Reviews distinguish such nonconscious processing from later conscious access and report [4], and separate global availability from metacognitive certainty [5]. Thalamocortical and corticocortical interactions contribute to state and content [16,21]. Distributed dynamic-core and global-workspace accounts do not require a single local inner observer [6].

Historical SAN (2022). The Whitepaper argued that the oscillator is the observer and that many coordinated cells collectively perform perception. It used canvas, ink, screen, and rendering analogies while explicitly denying that a literal LED screen exists in the brain.

Current SAN hypothesis. The observer operation is the state-changing relation between distributed ongoing activity and structured differences. Let the population state be

$$\mathcal{S}(t) = \left(\{s_i(t), \theta_i(t), \alpha_i(t), f_i(t)\}_{i=1}^N, \{\Gamma_{ij}(t)\}_{i,j=1}^N, b(t) \right), \quad (6)$$

where θ , α , and f denote measured phase, amplitude, and frequency variables; Γ_{ij} denotes effective interaction; and $b(t)$ denotes body state. A tonic field is not one universal frequency. It is the receiver-relative ongoing context against which a structured departure is measured. A phasic event is a change that alters one or more downstream state transitions.

The observer operation is defined functionally:

$$\mathcal{O}_t : (\mathcal{S}(t), X(t)) \mapsto (\mathcal{S}(t+1), d_{\text{out}}(t+1), a_{t+1}), \quad (7)$$

where X is incoming evidence, d_{out} is a declared later discrimination or report, and a is action. There is no additional variable for a hidden spectator.

This formulation preserves the stronger field hypothesis. Neural activity produces electrical, chemical, mechanical, metabolic, and thermal changes. Some electrical field effects can influence nearby neural activity under measured conditions [1,9,20]. SAN asks whether a jointly measured volumetric state contributes predictive and causal information beyond conventional spike, synaptic, and network variables. It does not assume that every field modality is equally informative, that a global electromagnetic lattice exists, or that field effects alone explain experience.

Engineering implication. Artificial observation should be attributed to the complete stateful perception-action system, not to a display buffer or a language module that narrates the result.

4.5.1 Recurrent thalamocortical concentration without a central controller

A task-relevant cortical pattern is vulnerable to distraction, competing representations, and internal noise. The systems problem is therefore not merely to create a pattern once. The nervous system must keep selected relations effective long enough for comparison, choice, action, or report while avoiding indiscriminate amplification of everything else.

Established thalamocortical physiology supplies a bounded mechanism family. In mouse mediodorsal-thalamus-prefrontal tasks, mediodorsal activity can sustain rule representations by amplifying functional connectivity without relaying the same categorical content [26]. Direction-specific perturbations show complementary roles: mediodorsal-thalamus-to-prefrontal projections support maintenance, whereas prefrontal-to-mediodorsal projections support subsequent choice [27]. Distinct mediodorsal pathways can amplify prefrontal signals when task inputs are sparse or suppress prefrontal noise when inputs are dense but conflicting [28].

In plain language, cortex can send a developing candidate state into a recurrent thalamic route; the thalamic circuit transforms which cortical relations remain effective; the returned influence changes the next cortical state; and repeated exchange can stabilize, sharpen, redirect, or terminate the candidate. The loop concentrates task relevance through selective recurrence rather than by copying the whole cortical pattern into a smaller structure.

SAN interprets this as a systems-level instance of the receive-transform-project cycle. Anatomical routing, inhibition, task state, and receiver condition determine whether a loop is available. SAN further proposes that phase-compatible timing may help open or reinforce a selected loop while incompatible timing and

inhibition help keep competitors separate. That timing claim is a discriminating SAN hypothesis; the cited thalamic studies establish amplification, maintenance, directional asymmetry, and signal / noise control, not the complete PWD mechanism.

The deep-learning analogy is selective pooling with feedback, but the identity is imperfect. Fixed pooling usually compresses a local array by a predetermined operation. A biological thalamocortical loop can change effective connectivity according to context, maintain relational content across repeated passes, and alter the cortical computation that generated its input. An artificial analogue would use a smaller recurrent control circuit to regulate a much larger high-dimensional workspace, with explicit safeguards against runaway gain, bottleneck distortion, and single-controller failure.

4.6 Stage 6: embodied action and returned consequence

Established. Vision, touch, and proprioception jointly influence bodily self-identification [3]. The free-energy framework treats action, perception, and learning as coupled optimization problems [8]. Those results provide the bounded multisensory and action-learning foundation used here; efference copy, vestibular input, and interoception are outside this paper’s evidentiary premises.

Historical SAN (2022). The Whitepaper traced information from sensory pathways through recurrent cortical and thalamic routes toward motor output. It treated bodily action as the large-scale output of a system that also receives the consequences of that output.

Current SAN hypothesis. Rendering is earned when the reconstructed variable changes prediction or action and is tested by returned consequences:

$$a_t = \pi(\hat{h}_t, b_t, g_t), \quad (8)$$

$$x_{t+1} = E(x_t, a_t, \eta_t), \quad (9)$$

$$e_{t+1} = d(\hat{x}_{t+1}, x_{t+1}), \quad (10)$$

where g_t is the current goal or value state, E is the body-environment transition, and e is a returned difference. SAN predicts that selected receiver-relative temporal and routing variables will improve the explanation of how that difference changes subsequent state.

Engineering implication. A self-model is incomplete if it cannot predict the consequences of action, compare them with returned evidence, and update future behavior.

4.7 Formal contract, units, and evidential status

Equations (1)-(10) are a typed systems contract. They specify which variables must be observed, estimated, or intervened on before the six-stage architecture can be tested. They are not yet one closed biophysical law. Each experiment must therefore declare the following roles rather than fit undefined symbols after seeing the result.

Symbol	Operational role	Type or units	Present evidential status
$x_i(t)$	input reaching receiver i	modality-specific measured vector	observed or preprocessed

Symbol	Operational role	Type or units	Present evidential status
$s_i(t)$	receiver’s active state	declared physiological or latent state	partly observed; model-dependent
m_i	learned or structural readiness	synaptic, intrinsic, morphological, or model parameters	partly observed; requires bounded assay
$c_i(t)$	context and modulation	task, neuromodulatory, inhibitory, or body-state variables	observed subset
$r_i(t)$	receiver-conditioned input event	output of declared reception model	estimated
$z_i(t)$	transformed local state	model-specific state vector	estimated and intervention-testable
$y_{i \rightarrow j}(t)$	target-specific output	spike, release, current, phase, or declared message variable	observed subset
$\hat{h}_t$	candidate reconstructed relation	identifiable latent variable or equivalence class	inferred; causal-use test required
a_t	action or policy output	categorical action, force, kinematics, speech, or control signal	observed
e_{t+1}	returned difference	units fixed by the declared discrepancy function d	computed from prediction and observation

The maps R_i , T_i , $P_{i \rightarrow j}$, G , π , E , and d must be specified before confirmatory evaluation. A fit is not evidence for the complete architecture when several maps can absorb the same unexplained variation. Nested comparators, parameter-identifiability analysis, held-out evaluation, selective perturbation, and matched rescue are therefore part of the formal claim rather than optional additions.

5. NAPOT as a multiscale hypothesis

The 2022 paper moved repeatedly among neurons, arrays, cortical columns, brain-wide activity, and artificial networks. The formal version must label scale explicitly.

5.1 Micro-level receive-transform-project motif

At the cellular level, one neuron can gather many synaptic inputs, transform them through dendritic, somatic, inhibitory, and intrinsic state, and project axonal outputs to many targets. This is an established receive-integrate-project motif. It is not, by itself, tomography by one neuron. Named feature streams and distant targets should be interpreted as examples distributed across pathways and populations, not direct connections of one generic cell.

5.2 Meso-level population reconstruction

At the local-circuit level, neurons and interneurons can form recurrent assemblies, compete, cooperate, synchronize, disperse, and produce population trajectories. A canonical laminar circuit can be used as one example, but cortical areas and species differ; no one column diagram should be presented as universal.

5.3 Macro-level distributed body-and-world model

At the systems level, established examples include hierarchies of cortical timescales [15], thalamocortical contributions to cortical state and conscious contents [16,21], engram-connectivity-dependent recall in a

mouse retrograde-amnesia model [17], and fronto-striatal coordination before vocal output [24].

NAPOT extends beyond any one of those results. It proposes that recurrent interactions among cortical, thalamic, hippocampal, basal-ganglia, cerebellar, brainstem, and peripheral systems maintain an integrated but differentiated body-and-world model whose consequences distribute into perception, memory, valuation, language, and action. That joined multiscale composition is a SAN hypothesis rather than an inference established by references [15-17,21,24].

5.4 When tomography is literal

The term *tomography* is literal only if five elements are specified:

1. a hidden state h_t to be reconstructed;
2. a forward projection model F relating h_t to partial observations $x_t^{(k)}$;
3. a reconstruction operator G ;
4. an identifiability or equivalence-class statement;
5. causal evidence that the biological system uses $\hat{h}_t$.

Formally,

$$x_t^{(k)} = F_k(h_t, u_t, \epsilon_t^{(k)}), \quad k = 1, \dots, K, \quad (11)$$

$$\hat{h}_t = G(x_{1:t}^{(1:K)}, s_t, M, A). \quad (12)$$

Decoder accuracy alone is insufficient. A researcher may decode information that the organism does not use. NAPOT requires intervention-sensitive evidence that changing the candidate reconstruction changes predicted behavior or internal processing while appropriate controls are held constant.

6. Phase-wave differentials and the tonic-phasic relation

The historical Whitepaper treated a deviation from ongoing oscillation as an information-bearing event. The current formulation separates established phase coding from the larger SAN operator.

Established. Spike timing relative to an ongoing rhythm can carry information beyond spike count in some preparations [14]. Different rhythms can define partly distinct functional interaction networks [19]. Neural coordination is heterogeneous across cell type, layer, area, state, and task.

Current SAN hypothesis. A Phase-Wave Differential is a bounded, receiver-relative, multidimensional difference between an expected ongoing state and an observed or transmitted event. One candidate typed event vector is

$$\mathbf{u}_{i \rightarrow j}(t) = [\phi, f, \alpha, \tau, q, v, \mathbf{x}, c]_{i \rightarrow j, t}, \quad (13)$$

where the components may denote phase, frequency, amplitude, duration, transmitted quantity, local variability, spatial or network location, and receiver context. The vector is a hypothesis space, not a declaration that every component is available or meaningful at every synapse.

The components cannot be added as raw numbers because they have different units and covariance. Any scalar magnitude or distance derived from the resulting differential must first declare a reference

distribution, scaling rule, missing-component policy, and covariance treatment. A whitened or Mahalanobis representation is one admissible choice; a raw Euclidean norm across phase, frequency, amplitude, duration, quantity, and location is not.

The tonic state supplies the current receiver-relative event reference:

$$\bar{\mathbf{u}}_j(t) = \text{Estimate}(\{\mathbf{u}_{k \rightarrow j}(t') : k \in \text{sources}(j), t' \in [t - w, t]\}), \quad (14)$$

and the candidate PWD vector is

$$\mathbf{p}_{i \rightarrow j}(t) = \mathbf{u}_{i \rightarrow j}(t) - \bar{\mathbf{u}}_j(t). \quad (15)$$

Coherence is not defined as universally beneficial. Near-synchronous activity can support recruitment or effective coupling; structured phase separation, inhibition, and splay-like relations can maintain boundaries, competition, or decoupling. The proposed content lies in the spatial pattern of coupling and separation, not maximal global synchrony.

6.1 How coordinated timing can accumulate effective drive for choice

A distributed candidate pattern still has to become a specific action. The problem is how many individually limited and noisy inputs can jointly bring a receiver or motor route toward threshold without assuming that synchrony is always beneficial or that measured population power is itself muscle energy. Inputs with compatible timing, geometry, and current sign can produce postsynaptic effects that sum at a receiver. Incompatible timing, inhibition, unfavorable geometry, or opposite current sign can weaken that sum. The resulting receiver-specific effective drive can therefore bias which candidate gains downstream control.

SAN calls a proposed signed, coherence-weighted estimate of this effective drive **electrical valence**. The name does not mean bulk net-charge accumulation, chemical valence, or psychological pleasantness. For candidate assembly c , a bounded estimator is

$$D_c(t) = \sum_i w_{ic} I_i(t) \cos(\phi_i(t) - \Phi_c(t)), \quad (16)$$

where I_i is a declared signed current or validated field proxy, w_{ic} includes geometry, projection efficacy, and receiver sensitivity, and Φ_c is the candidate reference phase. An electrical component of the candidate PWD is

$$\text{PWD}_{e,c}(t) = D_c(t) - \bar{D}_c(t). \quad (17)$$

The units of D_c follow the declared proxy. If I_i is current and w_{ic} is dimensionless, D_c has current units. If the inputs are voltage, field, current-source density, or normalized features, the paper must provide the corresponding forward model or standardization. Signals with different physical units must not be combined in the same sum merely because they share a phase.

Micah Blumberg's hypothesis is that coincident or phase-coherent recruitment can increase this effective drive when source geometry and current sign permit summation. The building PWD then contributes to which candidate assembly gains downstream control. Psychological valence remains a different variable: learned appetitive or aversive consequence can bias candidate recruitment, while electrical valence names the proposed physical drive of the recruited state.

Two task families make the choice claim concrete. **Represented-alternative selection** asks how concurrently considered options recruit competing association and action assemblies. **Learned-loop interruption** asks how an already prepared cue-prediction-action sequence becomes available to awareness, inhibition, delay, or redirection. SAN proposes a common candidate-to- action architecture, but the tasks require different controls. Reports that meditation, yoga, or other awareness practices increase choice are hypotheses about access and control; they do not by themselves establish the electrical-valence or PWD mechanism.

Bat vocalization provides a bounded example. Pre-vocal beta and high-gamma patterns in frontal cortex and dorsal striatum predicted echolocation versus communication in Weineck et al. (2020) [24]. The caudate condition contrast showed beta higher for echolocation and gamma higher for communication, while frontal high gamma was prominent before echolocation. This is a distributed, region-specific opponent pattern, not proof of one global beta-gamma anti-correlation. Garcia-Rosales et al. (2022) then found transfer-entropy-inferred frontal-to-auditory directionality before vocalization and auditory-to-frontal directionality during the post-vocal echolocation period [25].

The proposed motor chain is transformed rather than electrically literal: candidate recruitment changes cortical and subcortical motor routes, brainstem or spinal motor-neuron activity, neuromuscular transmission, muscle calcium, agonist-antagonist recruitment, force, and movement. For vocalization this includes laryngeal motor pathways. The bat studies did not record every link to the vocal folds and remain correlational for action initiation.

The specific test is whether coherence-weighted drive and $PWD_{e,c}$ predict and causally alter candidate commitment, motor output, and returned consequence beyond rate, power, anatomy, movement preparation, body state, area, depth, waveform, aperiodic activity, and flexible recurrent models. Condition-level beta-gamma opposition does not establish negative trialwise correlation; that relation must be measured directly.

6.2 What the Phase-Wave Differential adds

An arriving neural event never reaches an empty receiver. It reaches a cell or population that already has membrane state, inhibition, synaptic history, current oscillatory relations, spatial position, and an expected range of activity. The explanatory problem is therefore not only *when did the source fire?* It is *what multidimensional difference did that event make to this receiver, along this route, under this context, and what downstream consequence followed?*

Several established terms answer important parts of that question, but they do not name the complete operation:

Existing construct	What it measures or explains	Additional operation named by PWD
Phase coding	Information carried by spike or event timing relative to a reference rhythm.	Makes the difference explicitly source-to-receiver and permits phase to be one component among frequency, amplitude, duration, quantity, variability, location, and context.
Traveling waves	Propagation of organized activity through space and time.	Asks how a local wave-related event differs from the receiving state and whether that difference changes a specified target; a traveling wave can exist without satisfying that receiver-relative consequence test.

Existing construct	What it measures or explains	Additional operation named by PWD
Prediction error	Discrepancy between an expected or predicted variable and an observation.	Allows a receiver-relative physical differential even when no semantic prediction-error variable has been established, while requiring the reference state and consequence to be declared.
Communication through coherence	Phase relations that change effective interaction between populations.	Includes both compatible summation and structured separation, and treats routing as one part of a multidimensional content-bearing event rather than equating coherence with success.
Coefficient of variation	A scalar statistic of dispersion relative to a mean.	Treats local variability as one possible component, not as the full vector or mechanism.

PWD is therefore the name for the joined operator only after its parts are understood: establish the receiver’s reference state; measure the event relative to that state; retain the physically meaningful dimensions rather than collapsing unlike units; identify the target-specific transformation; and test whether the resulting vector predicts or causes a later state transition. The distinction matters experimentally. A preparation may exhibit phase coding, a traveling wave, coherence, or prediction error while the full receiver-relative vector adds no held-out or intervention value. Conversely, a target-specific multidimensional differential may add value even when no single comparator supplies a complete description. The dedicated PWD paper develops the calculus; this systems whitepaper specifies the role PWD must earn inside SAN.

7. Memory scaling without literal playback

The historical phrase *memory scaling* can be preserved if three distinct claims remain separate.

First, learning changes future sensitivity. Structural and physiological changes alter which cues recruit a cell or circuit.

Second, a local cue can initiate a larger distributed response. This is a recruitment claim, not proof that the local structure stores every detail of the later response.

Third, the distributed response can reconstruct relations absent from the cue. This is a completion claim and requires comparison with attractor, engram, latent-state, and sequence models.

Let c be the cue, $\hat{h}$ the candidate reconstruction, H the task-relevant hidden structure, and s_t the conditioning state. Define cue information, reconstruction information, and cue-conditioned added relational information as

$$\begin{aligned}
I_c &= I(c; H \mid s_t), \\
I_{\hat{h}} &= I(\hat{h}; H \mid s_t), \\
I_{\text{add}} &= I(\hat{h}; H \mid c, s_t).
\end{aligned} \tag{18}$$

A reconstruction is scientifically interesting when $I_{\text{add}} > 0$ on held-out data and when targeted perturbation changes the added relation rather than merely reducing global performance.

This formulation also bounds the idea that a dendrite could store a three-dimensional image. A dendritic branch may contribute high-dimensional, nonlinear, history-dependent basis functions to distributed

reconstruction. It should not be described as losslessly storing an arbitrary voxel array. Capacity depends on the number and diversity of effective synaptic variables, branch interactions, temporal state, population redundancy, and the reconstruction prior. The decisive question is how much task-relevant three-dimensional structure can be reconstructed above controls, not how many pixels can be assigned to one cell.

8. The distributed observer

The Whitepaper’s most important philosophical claim is also its most easily misunderstood: there is no little viewer watching a neural rendering. The distinction between distributed processing, access, report, and phenomenal claims is central to neighboring consciousness frameworks [4-6,21].

SAN proposes that the network itself performs the observation. Cells and circuits are the internal receivers and transmitters. Their coordinated activity changes the physical and functional state of the same system that discriminates, values, remembers, speaks, and acts. The aggregate observer operation is therefore not placed inside another hidden agent.

Three distinctions are required.

Observer operation versus observer report. A verbal report is one downstream consequence. It is not identical to the entire perceptual operation.

Access versus experience. Early decoding, recurrent stabilization, task access, confidence, report, no-report proxies, and phenomenology are different endpoints. A model should predict them separately.

Physical field versus explanatory field. The nervous system has measurable electrical, chemical, mechanical, metabolic, and thermal state variables. Calling their joint state a field does not show which variables are causal or sufficient for experience. SAN must earn the additional field claim through measurement and intervention.

The no-homunculus criterion can be stated as a graph property. If every content-bearing node requires another node that reads its output as a complete representation, the explanation regresses. For each content claim under test, a valid distributed account instead identifies at least one intervention-supported causal path from a target-specific output through later system state to a preregistered outcome:

$$\forall \chi \in \mathcal{C}_{\text{tested}}, \quad \exists \mathcal{P}_\chi = (i_0, \dots, i_K) \subseteq E_{\text{causal}} : \text{do}(y_{i_0 \rightarrow i_1} \leftarrow y') \Rightarrow \Delta s_{i_1} \Rightarrow \dots \Rightarrow \Delta o_\chi, \quad (19)$$

where o_χ is the declared behavioral, report, or internal-processing endpoint. This does not require every node to alter behavior or report. It requires the claimed content-bearing operation to participate in a testable causal path without adding a privileged terminal node whose sole role is to witness the rest.

8.1 Formal thalamocortical recurrence and concentration test

Let $\mathbf{c}_t^{(k)}$ be the cortical task-state estimate and $\mathbf{m}_t^{(k)}$ the state of a specified mediodorsal-thalamic circuit after recurrent pass k . A minimal directional loop is

$$\left(\mathbf{m}_t^{(k+1)}, \mathbf{c}_t^{(k+1)} \right) = \left(F_M \left(\mathbf{m}_t^{(k)}, \mathbf{c}_t^{(k)}, g_t \right), F_C \left(\mathbf{c}_t^{(k)}, \mathbf{m}_t^{(k+1)}, x_t \right) \right). \quad (20)$$

where g_t is task context and x_t is continuing evidence. F_M and F_C must be fitted as directional, intervention-testable maps rather than one undifferentiated recurrent network. Phase, inhibition, anatomy, and receptor state may enter either map when measured.

A concentration claim can then be tested with a dimensionless task-signal ratio

$$\kappa_K = \frac{\text{SNR}(\mathbf{c}_t^{(K)}; y_t)}{\text{SNR}(\mathbf{c}_t^{(0)}; y_t)}, \quad (21)$$

where y_t is a preregistered task-relevant variable and the signal / noise decomposition is fixed before evaluation. SAN predicts $\kappa_K > 1$ only for selected states and operating ranges, with direction-specific perturbations producing different maintenance and choice deficits. This is selective concentration, not creation of energy or information from nothing: recurrent biological work consumes metabolic resources, can amplify both signal and error, and can fail through runaway gain, distortion, or over-suppression. The strongest SAN result would show that receiver-relative phase and PWD variables improve this causal directional model beyond rate, power, anatomy, task, and flexible recurrence alone.

9. How distributed neural subsystems become one acting organism

The 2022 Whitepaper did not end with visual perception. It followed signals into thought, speech, movement, and returned sensory consequences. That agentic structure should remain central.

An agent is defined minimally as a bounded system with state that selectively transforms input, changes an output or environment, and changes future transformation through feedback. Nested biological subsystems can satisfy parts of this contract without becoming miniature conscious selves. The organism-level agent is the closed body-brain-action-consequence system.

SAN calls this joined organization the **Agentic Brain** hypothesis. It adds three claims:

1. local subsystems contribute different estimates, transformations, and control signals;
2. competition, inhibition, recurrent exchange, and state-dependent routing select temporary functional assemblies;
3. the resulting action and returned consequence update the same distributed system.

This creates a bridge to artificial multi-agent systems. Specialized artificial agents can share project state, goals, object identity, provenance, and action consequences while retaining distinct permissions and interfaces. That engineering architecture can be evaluated without claiming consciousness. A separate agentic-systems paper should develop the shared-context artificial case in detail.

10. Relation to neighboring theories

SAN is a synthesis with additional commitments. Similar vocabulary does not establish equivalence, and divergence in vocabulary does not by itself remove an operational match. The correct comparison is atomized by causal role.

Framework	Shared ground	What SAN adds or requires	Live alternative to SAN
Dendritic integration	Dendrites perform nonlinear, compartment-dependent integration.	Receive-transform-project across nested scales and receiver-relative temporal output.	Dendritic computation plus ordinary recurrent networks may explain the data without NAPOT.

Framework	Shared ground	What SAN adds or requires	Live alternative to SAN
Engram and attractor models	Learning changes distributed ensembles; partial cues can recruit memories.	Explicit scaling / reconstruction route connected to PWD, ongoing state, and action.	Pattern completion and dynamic engrams may explain recall without tomography.
Predictive processing [2,8]	Prior state and expected input shape processing; errors update models.	Typed physical differences, target-specific projection, and a distributed observer / action loop.	Hierarchical Bayesian or predictive-processing models may explain the same endpoints more simply.
Recurrent processing	Feedback and local recurrence can stabilize perceptual organization.	Multiscale reconstruction, receiver-relative phase relations, memory scaling, and embodied action.	Recurrent sensory processing may suffice without a global SAN field.
Global neuronal workspace	Selected content can become broadly available through recurrent amplification.	No single broadcast is assumed to be the observer; SAN separates access, confidence, report, and phenomenology.	Workspace ignition may explain access and report without NAPOT.
Communication through coherence	Relative phase can influence effective communication.	Coherence and structured dispersion jointly sculpt recruitment and boundaries; PWD is multidimensional.	Rate, anatomy, task state, or communication subspaces may explain routing.
Thousand Brains and reference-frame models [10]	Distributed cortical populations can learn object-centered relations and reach consensus across columns.	Phase-conditioned receive-transform-project cycles and an explicit observer / action hypothesis.	Reference-frame consensus may explain object knowledge without oscillatory tomography.
Dynamical-systems neuroscience	Population trajectories and latent states can explain behavior without fixed feature labels.	A named reconstruction burden, typed receiver-relative differences, and scale-linked causal tests.	Low-dimensional dynamics may absorb the SAN variables with no residual advantage.
Field and ephaptic accounts	Endogenous fields can modulate neural activity under bounded conditions.	A multimodal volumetric state connected to content-specific reconstruction and action.	Synaptic and network interactions may explain all relevant effects; ephaptic effects may remain small or local.

11. Predictions

The whitepaper makes the following separable predictions.

Prediction 1: receiver-relative temporal variables add information

For selected pathways, a model containing receiver-relative phase, burst, duration, release, and context variables will predict downstream state better than models using presynaptic rate and anatomy alone. Failure across appropriately powered preparations weakens the full PWD extension but not ordinary dendritic or recurrent processing.

Prediction 2: partial cues recruit added relational structure

A partial cue will recruit a distributed state containing task-relevant relations not present in the cue. Perturbing the proposed recurrent route will selectively remove those added relations. Nonspecific performance loss is insufficient.

Prediction 3: coherence and structured dispersion have complementary roles

Perceptual organization will depend on reproducible spatial patterns of locally coherent assemblies separated by structured phase dispersion or inhibition. Maximal global coherence will not be the optimal success condition.

Prediction 4: action-use distinguishes biological reconstruction from analyst decoding

A decoded internal variable qualifies as a candidate endogenous rendering only if selective perturbation changes a predicted discrimination, confidence judgment, action, or update and a matched rescue restores it.

Prediction 5: scale transitions preserve relations, not exact microstates

Micro-, meso-, and macro-level representations will share recoverable relational structure while differing in participating units and exact temporal realization. Exact cell-by-cell playback is not required.

Prediction 6: multimodal state improves content prediction only in bounded contexts

Joint electrical, hemodynamic, metabolic, mechanical, or thermal measurements may improve prediction of selected state transitions. The contribution must be assessed incrementally. A correlation between a field measure and task state does not establish causal observation.

Prediction 7: access, confidence, and report dissociate

SAN variables that predict early content identity will not necessarily predict stabilization, confidence, explicit report, or phenomenology. A successful model must declare the endpoint it predicts.

Prediction 8: artificial implementations benefit from shared persistent state

Artificial systems with receiver-specific state, recurrent reconstruction, embodied consequence, and shared project memory will show lower state-state error and better interruption recovery than isolated or summary-synchronized agents. This is an engineering prediction, not a machine-consciousness test [5].

Prediction 9: coherence-weighted drive links candidate state to motor consequence

In a bounded action-selection preparation, a candidate-specific coherence-weighted electrical- drive term will add held-out prediction of commitment, muscle or motor output, and returned consequence beyond matched rate, power, anatomy, movement, and recurrent-state models. A condition-level beta-gamma contrast will not be treated as trialwise anti-correlation without a direct test.

Prediction 10: thalamocortical recurrence selectively concentrates task signal

During a maintenance-and-choice task, repeated mediodorsal-thalamic and prefrontal exchange will improve a preregistered task-signal ratio over the initial cortical state within a bounded operating range. Perturbing thalamus-to-cortex and cortex-to-thalamus directions will yield different maintenance and choice deficits. Phase-aware variables qualify as a SAN-specific gain only if they add held-out and intervention value beyond rate, power, anatomy, uncertainty, and flexible recurrent comparators.

12. Experimental program

12.1 Experiment A: cellular receive-transform-project

Record identified presynaptic events, dendritic voltage or calcium, somatic spikes, terminal calcium or release, and postsynaptic response in a bounded circuit. Compare nested models:

1. presynaptic rate only;
2. rate plus anatomy;
3. phase and burst variables;
4. dendritic and receiver state;
5. full candidate PWD vector.

Use interventions that alter one component while preserving others as closely as possible. The experiment tests whether the joined receive-transform-project chain adds predictive and causal value.

12.2 Experiment B: partial-cue distributed reconstruction

Train an animal on a three-dimensional, multisensory, action-dependent scene. Present partial visual, tactile, or auditory cues while recording distributed populations. Quantify whether the recruited state recovers held-out object, location, body, or action relations. Perturb hippocampal, thalamic, cortical, or inhibitory routes based on explicit competing models. A successful SAN result requires content-specific loss and component-matched rescue.

12.3 Experiment C: external 3D capture to biological state

Create a synchronized external scene model using calibrated cameras, depth sensing, motion capture, and body pose. Record the participant with anatomically constrained fNIRS or fMRI, EEG or MEG, eye tracking, peripheral physiology, and, where ethically and clinically available, intracranial or laminar measurements. The external model supplies ground-truth geometry and event timing. The neural analysis asks which relational scene variables are reconstructable, when they become action-usable, and whether targeted noninvasive perturbation changes the same variables.

Ultrasound stimulation and optical or holographic measurement concepts require separate safety, dosimetry, spatial-resolution, and regulatory protocols. They must not be shown as routine clinical capabilities. The current experiment can begin with noninvasive observation and only add perturbation after an independently reviewed protocol.

12.4 Experiment D: content, access, confidence, report

Use masking, attention manipulation, matched-performance confidence conditions, and no-report proxies. Measure early sensory decoding, recurrent stabilization, task access, confidence, explicit report, and action separately. Test whether SAN variables predict distinct endpoints and whether a perturbation changes one endpoint while sparing others.

12.5 Experiment E: artificial reconstruction and shared agency

Implement the same receive-transform-project interfaces in an artificial embodied agent. Compare feedforward, recurrent, predictive-processing, latent-state, and SAN-inspired models on partial-cue completion, action consequence prediction, state continuity, and interruption recovery. Architectural success

does not establish sentience. A separate consciousness assessment would require an independently justified measurement theory.

12.6 Experiment F: candidate drive, motor realization, and returned consequence

In a vocal or limb-choice task, record candidate-related population activity, phase, power, current or validated field proxies, descending motor activity, EMG, force or kinematics, and returned sensory evidence. Compare rate-only, power-only, phase-only, current-only, geometry-free, coherence-weighted, and flexible recurrent models. Perturb the proposed timing relation while matching mean power and movement energy where feasible. Require a component-specific loss and a preregistered rescue. The bat data motivate the design but do not count as its confirmatory test.

12.7 Experiment G: directional thalamocortical concentration

Use a delayed rule, working-memory, or uncertainty task with simultaneous recordings from a defined prefrontal population and mediodorsal-thalamic projection classes. Estimate the directional maps in Equation (20), the concentration ratio in Equation (21), and receiver-relative phase / PWD variables before perturbation. Compare cortex-only, feedforward, undirected recurrent, rate-and-power, uncertainty-aware, and full directional SAN models. Perturb thalamus-to-cortex and cortex-to-thalamus pathways separately, then apply a matched temporal or pathway-specific rescue. A successful result requires selective recovery of the preregistered relation rather than nonspecific arousal or performance restoration.

13. Analysis and rejection criteria

Every experiment should preregister:

- the target variable and scale;
- the forward and inverse models;
- the comparator models;
- the training and held-out partitions;
- uncertainty and identifiability assumptions;
- perturbation targets and rescue conditions;
- the endpoint: decoding, access, confidence, report, action, or phenomenology;
- the result that would weaken or retire the SAN-specific claim.

The full NAPOT model should be rejected for a tested scope when all of the following hold:

1. the hidden state is non-identifiable beyond a broad equivalence class;
2. SAN variables add no held-out performance over well-tuned comparators;
3. selective perturbation produces no predicted content-specific effect;
4. rescue does not restore the predicted relation;
5. any decoded variable is not used by the organism's later processing or action.

Failure of literal tomography does not erase established dendritic integration, recurrent processing, memory completion, oscillatory coding, or embodied action. It retires the stronger conjunction.

14. Historical and updated figures

This public-safe current manuscript does not redistribute the six historical image binaries. Each figure section instead preserves the exact original filename, source date, Git commit, SHA-256, historical caption,

and public Whitepaper route. The images document what the public 2022 Whitepaper proposed; they are not medical evidence. Six medically reviewed counterparts are specified below as requirements for a later illustrated edition, but they are not represented as present in this manuscript. Historical and future figures have different jobs: the historical artifact preserves provenance, while any accepted modern counterpart must show anatomy, variables, boundaries, alternatives, and a decisive test.

Figure Pair 1: institutional and neurophysics frame

Public-safe provenance record. Historical binary omitted. Original filename: My\project-1\2).png; source timestamp: 2022-08-22\11:39:24\07:00; Git commit: 4902c64086e20df8cd8d06e598291aedd9808c34; SHA-256: 4DE2E59A10BE34AF4AB00272B5AFA4C9BE97177B40E0679CDA5DE810CFC760FE. The public historical sequence remains viewable at <https://github.com/v5ma/selfawarenetworks/wiki/Whitepaper>.

Historical caption. Original image used in the August 2022 SAN Whitepaper to frame SAN as a neurophysics, neural-computation, and artificial-neurology program. It is included as authorial provenance, not medical evidence.

Updated counterpart requirement. A clean program map connecting cellular physiology, local circuits, distributed systems, embodied action, measurement, falsification, and artificial implementation. It must not imply that one image proves a complete theory.

Figure Pair 2: reception and projection

Public-safe provenance record. Historical binary omitted. Original filename: 2D3D3A71-81A5-4342-A981-D13EFCD2CE41.png; source timestamp: 2022-08-22\21:57:05\07:00; Git commit: 6cd4d9e05f988824ad6e1fe8bf0f91f911313085; SHA-256: 4842E91903D6DA6F75490EABBA493CCB40C6972EA03454D81D8540DF2F1469F. The public historical sequence remains viewable at <https://github.com/v5ma/selfawarenetworks/wiki/Whitepaper>.

Historical caption. Original camera, microphone, pixel, and speaker analogy for dendritic reception and axonal projection. The analogy was intended to explain internal capture and transmission roles without a literal camera, screen, or inner viewer.

Updated counterpart requirement. A medically accurate pyramidal-neuron and local-circuit plate showing synapses on basal and apical dendrites, dendritic integration, axon initial segment, myelinated or unmyelinated axonal segments as appropriate, target-specific boutons, and multiple receiving cells. Established physiology and the candidate SAN receive-transform-project interpretation must be visibly separated.

Figure Pair 3: structural memory and learned sensitivity

Public-safe provenance record. Historical binary omitted. Original filename: My\project-1\9).png; source timestamp: 2022-08-22\12:11:50\07:00; Git commit: ef83d2f65163a50805d7a953071b6e7461e3e0e9; SHA-256: 9485FD3F6F424E83AA207ECB6EE15929B35CA29A921A849169D4E85E3611DF21. The public historical sequence remains viewable at <https://github.com/v5ma/selfawarenetworks/wiki/Whitepaper>.

Historical caption. Original illustration of the claim that changed synapses and cell morphology alter future pattern sensitivity.

Updated counterpart requirement. A before-learning, after-learning, partial-cue, and reconstructed-population sequence. It must distinguish spine and synaptic plasticity, intrinsic excitability, ensemble connectivity, cue-driven recruitment, and behavior. A dendrite is changed readiness, not a complete stored picture.

Figure Pair 4: recursive scaling and asymmetric reinstatement

Public-safe provenance record. Historical binary omitted. Original filename: My\project-1\5.png; source timestamp: 2022-08-22\11:39:24\07:00; Git commit: 4902c64086e20df8cd8d06e598291aedd9808c34; SHA-256: CB3E6BF34F89C543F286A8CD879889F440830A96A5F59B21F3008BD62A3CFCE0. The public historical sequence remains viewable at <https://github.com/v5ma/selfawarenetworks/wiki/Whitpaper>.

Historical caption. Original FRACTAL image for recursive cortical scaling and asymmetric reinstatement across levels.

Updated counterpart requirement. Three explicitly labeled scales: micro receive-integrate-project, meso recurrent local circuit, and macro distributed thalamocortical / hippocampal / association network. Relations may recur across scales, but the anatomy and state variables must differ. The figure must not depict exact geometric self-similarity as established biology.

Figure Pair 5: oscillatory self-conception and distributed observation

Public-safe provenance record. Historical binary omitted. Original filename: 330CE863-0AEF-4749-9C27-BE4D8CECC3F2.png; source timestamp: 2022-08-22\20:06:45\07:00; Git commit: 3ed84b2ad4b8c2e7632cd7547e0ae13a9215dd6d; SHA-256: F80FC91DE6AAB4AE70476383C09BFE2A74901ECDEDEF428E63EDACA7676D6AFE. The public historical sequence remains viewable at <https://github.com/v5ma/selfawarenetworks/wiki/Whitepaper>.

Historical caption. Original image for the claim that coordinated oscillating populations collectively perform observation and self-conception.

Updated counterpart requirement. A no-homunculus distributed-state plate showing tonic context, structured phasic changes, local coherence, structured dispersion, thalamocortical recurrence, body-state input, and separable consequences for discrimination, confidence, report, and action. No cortical region may be labeled the viewer or consciousness center.

Figure Pair 6: Artificial Neurology

Public-safe provenance record. Historical binary omitted. Original filename: My\project-1\6.png; source timestamp: 2022-08-22\11:39:24\07:00; Git commit: 4902c64086e20df8cd8d06e598291aedd9808c34; SHA-256: A843DA9C32E8CD44856FDE4964765B658EAC0785C3C2CA81BF73D738D708554F. The public historical sequence remains viewable at <https://github.com/v5ma/selfawarenetworks/wiki/Whitpaper>.

Historical caption. Original Whitepaper transition from biological SAN to Artificial Neurology and the construction of artificial minds.

Updated counterpart requirement. A two-column biological-to-artificial translation. Each artificial component must inherit a bounded operation rather than a decorative brain analogy: role-conditioned reception, persistent state, target-specific projection, recurrent reconstruction, embodied consequence, provenance, and explicit consciousness boundary.

No updated counterpart is accepted until its anatomy, mechanism, labels, claim boundaries, and rights status are reviewed individually. Rejected variants remain quarantined and are not eligible for publication.

15. Source genealogy

The present manuscript is a new synthesis. Its chronology separates an early functional ancestor, a later mechanical expansion, the named 2022 architecture, and subsequent formalization. A source can establish authorship history without establishing that the biological hypothesis is correct.

Date or period	Exact source role	Atom or architecture preserved here	Provenance and boundary
2011-08-24 / 08-28 owner-reported; 2025-12-28 public fixation	<i>The Self Is a Prediction</i> dialogue [29]	Self as a reinforced prediction pattern; whole-neocortex / whole-process framing; world as internally constructed prediction pattern	The owner-supplied description and reported Studio metadata assert recording on August 24, 2011 and upload on August 28, 2011; no preserved Studio export or screenshot is currently packaged. The verifiable public fixation is December 28, 2025. This is an authorship-genealogy anchor, not adversarial proof of public availability in 2011 and not mature NAPOT or PWD.
2012-07-30 local / 07-31 UTC	<i>Happiness Makeover with Brainwaves Orientation</i> [30]	Hofstadter-style feedback loop; self-awareness as engineered recursive feedback; synchronized EEG / video coincidence protocol; Hawkins prediction lineage	Public YouTube source. It establishes operational ancestry, not the later multidimensional PWD equation.
2012-08-01 local / 08-02 UTC	<i>Neo Mind Cycle and My Happiness Makeover</i> [31]	EEG-driven light / sound return as a two-way sensory-neural conversation; coordinated firing described as an oscillatory / electromagnetic carrier	Public YouTube source. It does not establish that every later field variable or frequency assignment is correct.
2012-08-23 local / 08-24 UTC	<i>Inner Peace</i> dialogue [32]	Mind as a projector predicting the world; internal model supplies most experienced detail while external evidence perturbs it	Live platform timestamp is preserved. This is a functional neural-rendering ancestor, not a backdated tomography term.
2017-04-11	Neural Lace Podcast founding episode [33]	Spatial computation in neuroscience; time-varying 3D neural matrix; thalamic convergence; bidirectional read / write BCI; Alternating Transmission Protocol; present-pattern versus desired-pattern differential	Public YouTube episode with exact transcript route. This is the best episode-level mechanical bridge between Neo Mind Cycle and named SAN.

Date or period	Exact source role	Atom or architecture preserved here	Provenance and boundary
2017-04-13 to 05-11	Neural Lace Podcast episodes 2-5 [38]	Dendrites as local computers; coincidence as a neural bit; broccoli recognition as a shared temporal-spatial pattern; volumetric / light-field rendering as a receive-transform-project analogy	Public media and transcript-family map. Speaker attribution remains claim-specific; external rendering is an operational analogy, not biological validation.
2017-06-11 to 2020-05-12	Remaining Neural Lace instrument, XR, embodiment, and biofeedback episodes [38]	EEG-XR feedback, embodiment, NerveGear, multimodal measurement, fNIRS / HEG / EIT tradeoffs, and continuity with 2012	All 11 tracked transcripts are summarized, but only n1p1.md appears in the current governed Git-tranche receipts. Ten tracked transcripts still require dedicated source-atom ledgers; the missing n1p6 label is a numbering gap, not evidence of a missing public link.
2022-06-08 Git fixation begins; public visibility self-reported no later than 06-17; exact PWD phrase fixed 08-23	Public SAN Git history [23, 38]	Neural oscillation tomography; distributed observer; receiver-defined temporal windows; array-to-array receive-transform-project; exact PWD naming	Repository commits begin June 8 local / June 9 UTC. The changelog self-reports first public sharing by June 17, but that changelog statement was itself Git-fixed on August 26. Exact atom claims use immutable commits in Supplement [38]; the exact PWD phrase appears in commit fc922f143b68f3c058e8fdc54ceed7b9d6a54749 on August 23, 2022. Earlier functional ancestors are not relabeled as if they already used the 2022 terms.
2022-07-28 to 09-18	Four public SAN / NAPOT video explainers [34-37]	Synapse-to-network memory scaling; distributed observer; selective reception and projection; tonic / phasic canvas; inhibition, choice, action, Fourier / tomographic and volumetric analogies	All 1,534 transcript segments were reviewed into 89 bounded atom families. This is complete source-body review of these four recordings, not 89 independent biological validations.
2022-08-22 to 08-28	SAN Wiki Whitepaper and six-image sequence [22]	Public six-stage systems narrative and visual architecture	The six images are exact authorial-provenance artifacts. Publication rights for embedded third-party material require separate clearance.
2023-2026	Later SAN papers, source recovery, Book 2, and formalization [38]	Typed PWD, explicit scale labels, identifiability, comparators, perturbation-rescue, endpoint separation, source-faithful correction	New detail retains its actual date. The broader Git corpus remains inventoried but not exhaustively human-atomized; 171 artifacts remain in the current semantic-review queue.

The immutable 2022 Whitepaper anchor is Wiki commit caf2326ac04477f4ebf1bc45f582849d258429e5.

The archived local whitepaperdraft3.md, SHA-256 8DFF765AF198D43A2B65CFDF28B2CDCB55D341BDFAD619BCC26F77849C9FF6D7, is a later donor snapshot first fixed in Git on December 13, 2023; it must not be presented as an independently fixed 2022 file. The paper-local Claim, Source, and Genealogy Atlas [38] binds immutable or date-qualified source routes, selected local hashes, source-review status, and claim destinations without modifying the upstream Book 2 or SAN Wiki genealogy files.

16. Limitations

SAN currently joins findings from several scales and literatures more strongly than any one experiment supports. Dendritic integration does not prove neural rendering. Phase coding does not prove PWD. Recurrent processing does not prove tomography. Engram activation does not prove lossless memory scaling. Endogenous field effects do not prove a conscious field. Successful artificial reconstruction does not prove machine consciousness.

The current equations define variables and experimental burdens. They are not a completed quantitative theory with uniquely estimated parameters. Different preparations may require different state spaces. Terms such as *tonic*, *phasic*, *field*, *array*, *rendering*, *projection*, and *observer* must always be tied to measured variables.

The theory also risks overgeneralizing from canonical cortical diagrams. Cortical areas, layers, cell types, species, thalamic nuclei, hippocampal circuits, cerebellar circuits, basal-ganglia loops, and peripheral pathways differ. A figure may illustrate one bounded circuit without becoming a universal template.

Finally, consciousness science lacks one accepted ground-truth measure of phenomenal experience. The paper therefore separates state, content, decoding, access, confidence, report, no-report proxies, action, and phenomenology. SAN should be judged against the endpoint it actually predicts.

17. Conclusion

The 2022 SAN Whitepaper proposed a coherent systems story: neural tissue receives partial patterns, transforms them through learned structure, projects changed signals, recruits distributed memory and oscillatory relations, and closes the cycle through action. Its most important claim was that the distributed system itself performs the observer operation. No additional inner viewer is required.

The present formalization preserves that story while replacing categorical declarations with explicit models and tests. NAPOT is now a candidate multiscale receive-transform-project architecture. Neural rendering is an action-usable recurrent reconstruction, not a literal screen. Memory scaling is cue-driven distributed reinstatement, not lossless playback from one dendrite. Oscillatory coherence and structured dispersion are candidate routing and differentiation variables, not universal markers of success. The field-cell proposal remains a measurable multimodal hypothesis, not a decorative metaphor and not an established theory of consciousness. Artificial Neurology remains an engineering destination whose capabilities and consciousness-relevant properties must be tested separately.

This makes the Whitepaper suitable for its intended role. It is the concise map of the SAN architecture, the focused papers develop individual mechanisms, and SAN Book 2 supplies the comprehensive historical, biological, technical, and philosophical account. The next step is not broader assertion. It is a sequence of medically accurate figures, preregistered comparator experiments, and independent scientific review capable of showing which parts of the architecture survive contact with data.

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Authorship and research history

The core ideas, scientific hypotheses, and source program in this whitepaper are Micah Blumberg’s work, developed across research conducted from 2011 through 2026. Substantial foundations were documented before ChatGPT became publicly available in November 2022. AI assisted the current edition with source retrieval, comparison, mathematical and coding formulations, grammar, formatting, and medical, biological,

and physics accuracy checks. It was not the origin or author of the SAN, NAPOT, PWD, neural-rendering, choice, or Artificial Neurology theories. Exact sources, equations, historical attributions, figure rights, and scientific claims remain subject to Micah's review and independent subject-matter review before public deposit.